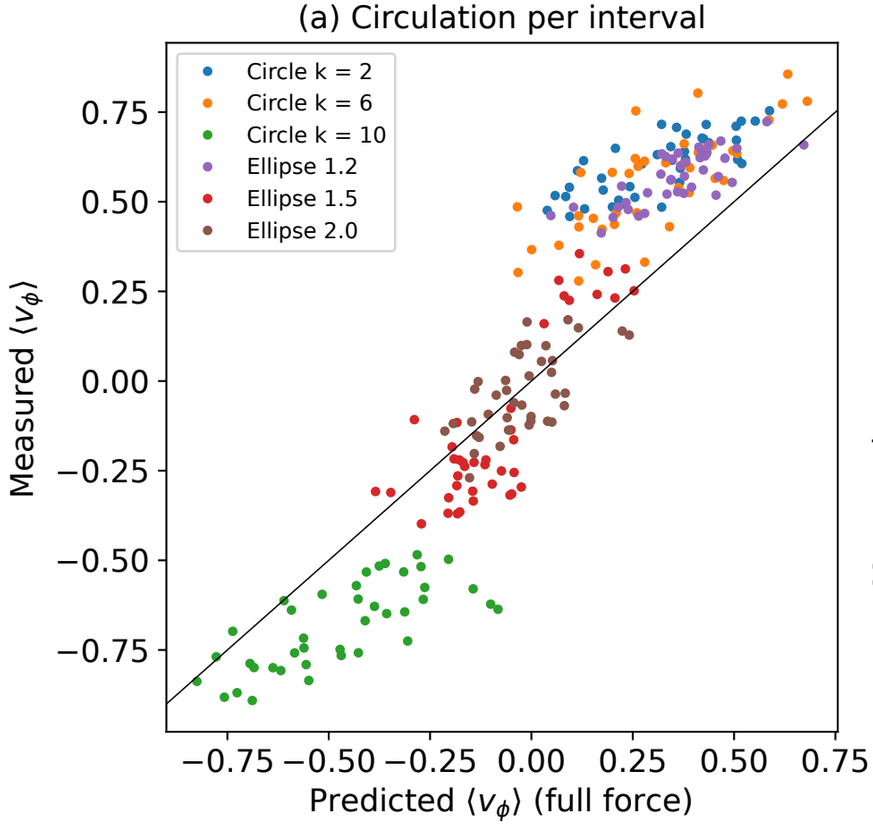
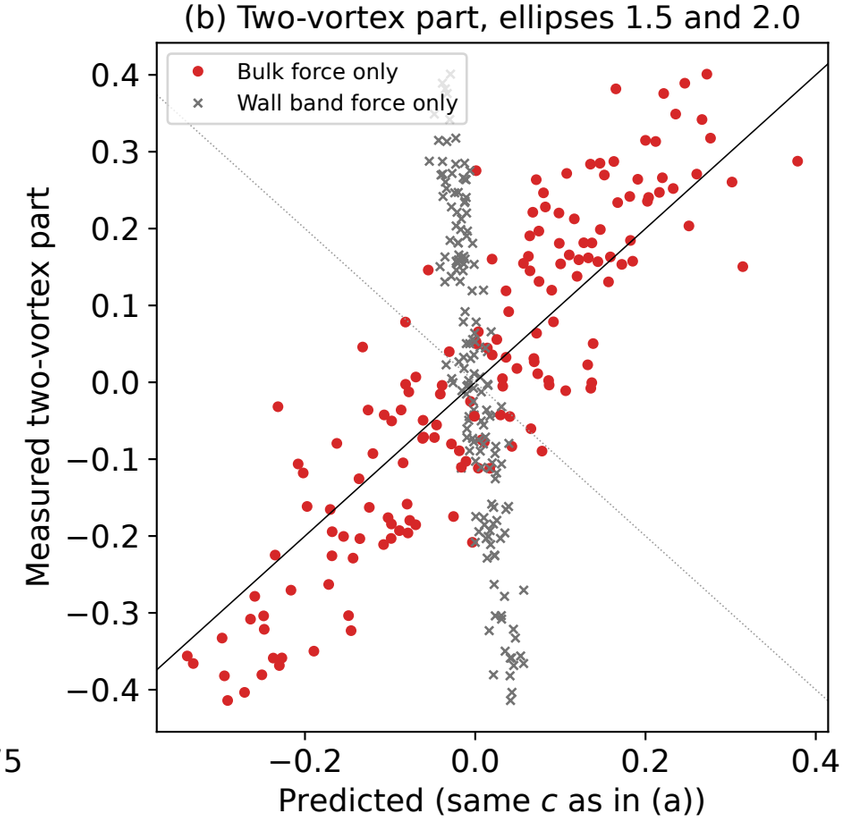
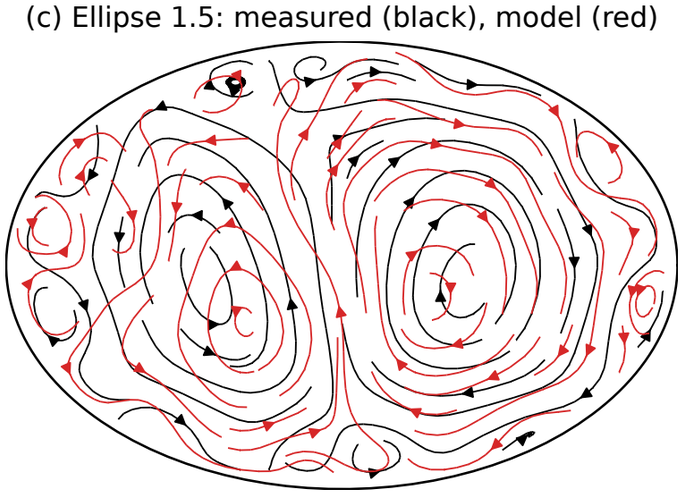

(a) Circulation per interval
Circle k = 2
Circle k = 6
Circle k = 10
Ellipse 1.2
Ellipse 1.5
Ellipse 2.0
Measured ⟨v_ϕ⟩
Predicted ⟨v_ϕ⟩ (full force)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


(b) Two-vortex part, ellipses 1.5 and 2.0
Bulk force only
Wall band force only
Measured two-vortex part
Predicted (same c as in (a))
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4


(c) Ellipse 1.5: measured (black), model (red)